

Light, Reflection and Refraction

Laws of reflection, images formed by concave and convex mirrors, laws of refraction, images formed by convex and concave lenses, and power of a lens.

Introduction

Light is a form of energy that helps us to see objects. It travels in a straight line. When light falls on a surface, it can be reflected or refracted. This chapter studies how images are formed by mirrors and lenses.

We use two new ideas: spherical mirrors (concave and convex) and spherical lenses (convex and concave). We learn how to draw ray diagrams and how to use formulas to find the position and size of images.

Reflection of Light and Laws

Reflection is the bouncing back of light when it hits a smooth surface.

Terms to know: the ray that falls on the surface is the incident ray. The ray that goes back is the reflected ray. The point where the ray hits the surface is the point of incidence. The line perpendicular to the surface at that point is the normal.

Angle of incidence (i) is the angle between incident ray and normal. Angle of reflection (r) is the angle between reflected ray and normal.

Laws of reflection

1. The incident ray, the reflected ray and the normal to the surface at the point of incidence all lie in the same plane.
2. The angle of incidence is equal to the angle of reflection. That is, $i = r$.

These laws are true for all mirrors, plane and spherical.

Spherical Mirrors

A spherical mirror is a part of a hollow sphere with one side polished and the other side reflecting.

- **Concave mirror:** reflecting surface is inside the curve. It is curved inward like a cave. It converges light.
- **Convex mirror:** reflecting surface is outside the curve. It is bulged outward. It diverges light.

Important terms:

- **Pole (P):** centre of the mirror surface.
- **Centre of curvature (C):** centre of the sphere of which the mirror is a part. Distance PC is radius of curvature R.
- **Principal axis:** straight line through P and C.
- **Focus (F):** point on principal axis where rays parallel to principal axis meet after reflection (concave) or appear to come from after reflection (convex).
- **Focal length (f):** distance between pole and focus. $f = R/2$.
- **Aperture:** diameter of the reflecting surface.

For a concave mirror, focus is in front of the mirror. So f is taken as negative. For a convex mirror, focus is behind the mirror. So f is taken as positive. This is as per the New Cartesian Sign Convention used in NCERT.

Images Formed by Concave and Convex Mirrors

We draw ray diagrams using two rules:

- A ray parallel to principal axis passes through focus after reflection in a concave mirror, or appears to come from focus in a convex mirror.
- A ray passing through focus becomes parallel to principal axis after reflection in a concave mirror. A ray going towards focus becomes parallel in a convex mirror.
- A ray passing through centre of curvature reflects back along the same path.
- A ray hitting the pole reflects with equal angle.

Concave mirror: position of object and image:

- Object at infinity: image at focus, highly diminished, real and inverted.

- Object beyond C: image between F and C, diminished, real and inverted.
- Object at C: image at C, same size, real and inverted.
- Object between C and F: image beyond C, enlarged, real and inverted.
- Object at F: image at infinity, highly enlarged, real and inverted.
- Object between P and F: image behind the mirror, enlarged, virtual and erect.

Convex mirror: image formation:

A convex mirror always forms a virtual, erect and diminished image behind the mirror, between P and F. The image gets smaller as the object moves away. Field of view is large. That is why convex mirrors are used as rear-view mirrors in vehicles and as vigilance mirrors in shops.

Uses of concave mirrors: as shaving mirrors (when face is between P and F, enlarged erect image), by dentists to see enlarged image of teeth, in torches and headlights as reflectors, and in solar furnaces to concentrate light.

Mirror Formula and Magnification

New Cartesian Sign Convention for mirrors:

- Pole P is taken as origin. Principal axis is X-axis.
- Incident light is taken from left to right.
- Distances measured in the direction of incident ray are positive, opposite are negative.
- Heights above principal axis are positive, below are negative.

So object distance u is always negative (object is in front of mirror). f is negative for concave mirror and positive for convex mirror. v is negative if image is in front, positive if behind.

Mirror formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

where f is focal length, v is image distance, u is object distance. All with sign as per convention.

Magnification by a mirror

$$m = h'/h = -v/u$$

where h' is image height and h is object height. If m is negative, image is real and inverted. If m is positive, image is virtual and erect. If $|m| > 1$, image is enlarged. If $|m| < 1$, image is diminished.

Worked Example

An object is placed 30 cm in front of a concave mirror of focal length 15 cm. Find the image distance.

Given: $u = -30$ cm, $f = -15$ cm. Use $1/f = 1/v + 1/u$.

$$1/(-15) = 1/v + 1/(-30)$$

$$1/v = -1/15 + 1/30 = -2/30 + 1/30 = -1/30$$

So $v = -30$ cm. Image is 30 cm in front of mirror, real, inverted, same size (at C). Magnification $m = -v/u = -(-30)/(-30) = -1$.

Refraction of Light and Laws

Refraction is the bending of light when it goes from one medium to another. It happens because light travels at different speeds in different media. Light travels fastest in vacuum (3×10^8 m/s). In glass it is slower.

When light goes from rarer to denser medium (air to glass), it bends towards the normal. When it goes from denser to rarer (glass to air), it bends away from the normal.

Laws of refraction

1. The incident ray, the refracted ray and the normal at the point of incidence all lie in the same plane.
2. Snell's law: the ratio of sine of angle of incidence to sine of angle of refraction is constant for a given pair of media. That is,

$$\sin i / \sin r = \text{constant} = n_{21}$$

This constant is called refractive index of second medium with respect to first.

Refractive index: $n = \text{speed of light in vacuum} / \text{speed of light in medium}$. Often written as $n = c/v$. For example, n of glass is about 1.5, n of water is 1.33. Absolute refractive index is with respect to vacuum.

Higher n means optically denser medium, but optically denser does not always mean physically denser.

Spherical Lenses

A lens is a transparent material with two spherical surfaces.

- **Convex lens (converging lens):** thicker in the middle, thinner at edges. It is of three types: double convex, plano-convex, concavo-convex. A convex lens converges light rays.
- **Concave lens (diverging lens):** thinner in the middle, thicker at edges. It is of three types: double concave, plano-concave, convexo-concave. A concave lens diverges light rays.

Terms for lenses:

- **Optical centre (O):** centre of the lens. A ray through O goes without deviation.
- **Centre of curvature:** centres of the two spheres forming the lens, C1 and C2. Radius R1 and R2.
- **Principal axis:** line through C1, C2 and O.
- **Focus (F):** point where rays parallel to principal axis converge (convex) or appear to diverge from (concave) after refraction. A lens has two foci, F1 and F2, one on each side.
- **Focal length (f):** distance between optical centre and focus.

For a convex lens, f is positive. For a concave lens, f is negative, as per sign convention.

Images Formed by Convex and Concave Lenses

Ray diagram rules for lenses:

- A ray parallel to principal axis passes through focus on the other side (convex) or appears to come from focus on the same side (concave).
- A ray passing through optical centre goes straight without bending.

- A ray passing through focus becomes parallel to principal axis after refraction.

Convex lens images:

- Object at infinity: image at F_2 , highly diminished, real and inverted.
- Object beyond $2F_1$: image between F_2 and $2F_2$, diminished, real and inverted.
- Object at $2F_1$: image at $2F_2$, same size, real and inverted.
- Object between F_1 and $2F_1$: image beyond $2F_2$, enlarged, real and inverted.
- Object at F_1 : image at infinity, highly enlarged, real and inverted.
- Object between O and F_1 : image on same side as object, enlarged, virtual and erect. This is used as a magnifying glass.

Concave lens images:

A concave lens always forms a virtual, erect and diminished image on the same side as the object, between O and F . Like convex mirrors, it has a wide field but always diminished.

Lens Formula, Magnification and Power

Sign convention for lenses: optical centre O is origin. Light comes from left. Distances in direction of incident ray are positive. So object distance u is negative. f is positive for convex lens, negative for concave lens. v is positive if image is on opposite side of object (real image, convex lens), negative if on same side (virtual image).

Lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Note the minus sign. This is different from mirror formula.

Magnification by a lens

$$m = \frac{h'}{h} = \frac{v}{u}$$

For lenses, $m = v/u$ (no minus sign as in mirrors). Positive m means erect and virtual. Negative m means inverted and real.

Power of a lens

$$P = 1/f$$

where f is in metres. Unit of power is dioptre (D). $1 \text{ D} = 1/\text{m}$. For a convex lens, P is positive; for a concave lens, P is negative.

Worked Example

A convex lens has focal length 20 cm. An object is placed 30 cm in front of it. Find image distance and magnification.

Given: $f = +20 \text{ cm}$, $u = -30 \text{ cm}$. Use $1/f = 1/v - 1/u$.

$$1/20 = 1/v - 1/(-30) = 1/v + 1/30$$

$$1/v = 1/20 - 1/30 = 3/60 - 2/60 = 1/60$$

So $v = +60 \text{ cm}$. Real image on the other side, 60 cm from lens. Magnification $m = v/u = 60/(-30) = -2$. So image is inverted and enlarged 2 times.

Worked Example

A concave lens has power -2 D. What is its focal length?

$$P = 1/f \quad \text{so} \quad f = 1/P = 1/(-2) = -0.5 \text{ m} = -50 \text{ cm}$$

So focal length is 50 cm, negative because lens is concave.

Summary

- Laws of reflection: $i = r$ and all three rays and normal are in same plane.
- Concave mirror converges, convex mirror diverges. $f = R/2$.
- Concave mirror forms real inverted images except when object is between P and F , where image is virtual and erect. Convex mirror always forms virtual, erect, diminished image.
- Mirror formula: $1/f = 1/v + 1/u$. Magnification: $m = -v/u$.
- Refraction is bending due to change in speed of light. Snell's law: $\sin i / \sin r = n_2/n_1$.
- Refractive index $n = c/v$. Higher n means optically denser.

- Convex lens converges, concave lens diverges. Convex lens f is positive, concave lens f is negative.
- Convex lens forms real inverted images except when object is between O and F , where image is virtual and erect. Concave lens always forms virtual, erect, diminished image.
- Lens formula: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$. Magnification: $m = \frac{v}{u}$.
- Power of lens: $P = \frac{1}{f}$ (f in metres). Unit is dioptre.

Practice Questions

1. State the laws of reflection and the laws of refraction. Write Snell's law.

2. An object is placed 20 cm in front of a concave mirror of focal length 15 cm. Find the position and nature of the image. Also find magnification.

3. Why is a convex mirror used as a rear-view mirror in vehicles? Give two reasons.

4. Draw ray diagrams for a convex lens when the object is (a) at $2F$, (b) between F and $2F$, (c) between O and F . State nature and size of image in each case.

5. A concave lens has focal length 25 cm. An object is placed 50 cm in front of it. Find image distance and magnification.

6. A concave lens has power -2 D. Find its focal length in metres and centimetres.